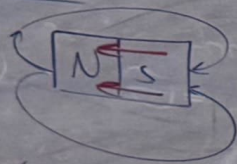


Magnetic field

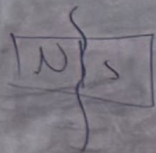
Magnetic field is a region around a moving charge

Electric field is a region around (moving - stationary) charge



B lines are directed from N into S (out of magnet)

B lines are directed from S into N (in magnet)



if we cut a magnet into two parts:

we get two smaller magnets

There is no isolated magnetic pole

↳ no single magnetic pole

There is a pair of magnetic poles

Monopole does not exist

Magnetic Force

$\theta = 90^\circ$
 $\vec{F}_B$ is perpendicular to $\vec{v}$ and $\vec{B}$.

$\vec{F}_B$ acts only on a moving charge in $\vec{B}$.

$\vec{F}_B$ does change a direction of $\vec{v}$.

$\vec{F}_B$ does not change a magnitude of $\vec{v}$ → magnitude

$\vec{F}_B$ is directed toward a center → velocity unchanged → $\vec{F}_B$ does no work

بسم الله الرحمن الرحيم

Parallel

anti parallel

$\ominus = \odot$, $\ominus = 180^\circ$

$\vec{F}_B = 0$

Straight motion

$0^\circ < \theta < 90^\circ$

$\vec{F}_B = qvB \sin \theta$

helix motion

$\theta = 90^\circ$

$\vec{F}_B = qvB \text{ max}$

Circular motion

On a moving charge:

$\vec{F}_B = q \vec{v} \times \vec{B}$

$F_B = qvB \sin \theta$

$\vec{F}_B = qvB \text{ max}$

On a wire has current:

$\vec{F}_B = I \vec{L} \times \vec{B}$

$F_B = ILB \sin \theta$

$\vec{F}_B = ILB \text{ max}$